Region	Quadrant	Nature of x,y	Signs of the coordinates
XOY	I	$x>0,\ y>0$	(+,+)
$X'OY$	II	$x<0,\ y>0$	(–,+)
$X'OY'$	III	$x<0,\ y<0$	(–,–)
XOY'	IV	$x>0,\ y<0$	(+,–)

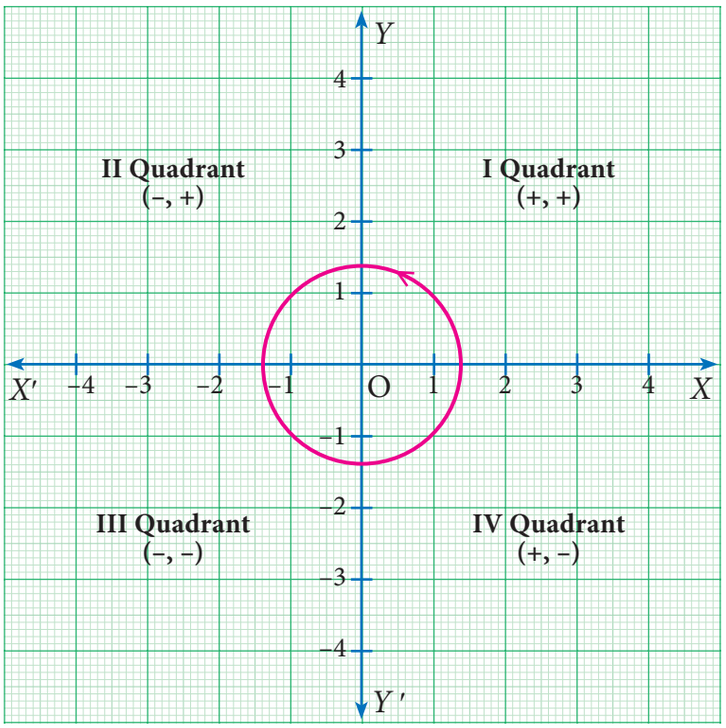


Fig. 5.7

Note

- For any point P on the x axis,the value of y coordinate (ordinate) is zero that is, P (x,0).
- For any point Q on the y axis, the value of x coordinate (abscissa) is zero. that is, Q (0, y)
- $(x, y) \neq (y, x)$ unless x= y
- A plane with the rectangular coordinate system is called the Cartesian plane.

5.2.1 Plotting Points in Cartesian Coordinate Plane

To plot the points (4, 5) in the Cartesian coordinate plane. We follow the x – axis until we reach 4 and draw a vertical line at $x = 4$.

Similarly, we follow the y – axis until we reach 5 and draw a horizontal line at $y = 5$.

The intersection of these two lines is the position of (4, 5) in the Cartesian plane.

This point is at a distance of 4 units from the y-axis and 5 units from the x-axis. Thus the position of (4, 5) is located in the Cartesian plane.

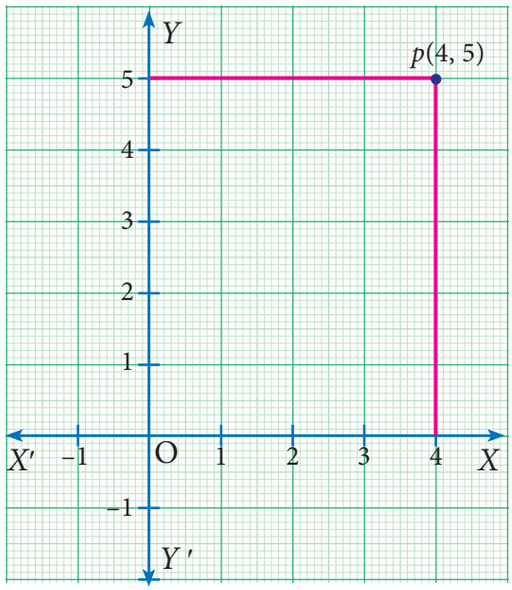


Fig. 5.8

5-Coordinat_1.indd 190 02/12/2022 21:44:42